Speech Recognition and Large Language Models for Monitoring Knowledge Acquisition from Teacher–Student Dialogue: A Literature Review

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***Abstract*—This review examines the evidence base for an intelligent system that estimates students' knowledge acquisition from recorded teacher–student dialogue in Kazakh and Russian. The system separates speakers, transcribes the lesson, uses a large language model (LLM) to judge the student's understanding of each topic, and reports the result to teachers, students, and parents. We analyzed 23 peer-reviewed journal articles (13 research studies and 10 reviews) on automatic speech recognition (ASR), speaker diarization, LLM-based dialogue analysis and answer grading, knowledge tracing, learning analytics, and explainable AI. Speech technologies are usable but task- and language-dependent: agreement between a speech-verification system and human raters ranged from $\kappa = 0.15$ to 0.70 across tasks, classroom diarization error was about 34%, and a Kazakh-adapted Whisper model still produced a word error rate of 26.3% on Kazakh against 11.9% on Russian read speech. LLMs agree with human coders at the category level ($\kappa = 0.69$) but not at the level of fine-grained codes, and an out-of-the-box LLM grades short written answers with F1 of 0.61–0.74. Explanations increase teachers' trust in AI analysis. No reviewed study connects these components end to end, validates them against subject knowledge, or evaluates spontaneous Kazakh, Russian, or code-switched dialogue. From these gaps we derive design and evaluation requirements for the proposed system.**

***Index Terms*—automatic speech recognition, speaker diarization, large language models, classroom dialogue, knowledge assessment, knowledge tracing, explainable AI, Kazakh, Russian.**

I. Introduction

WHAT a student says during a lesson is often the earliest evidence of whether a topic has been understood. Answers, hesitations, repeated questions, and corrections reveal misconceptions long before a test does. Teachers rarely have time to review lesson recordings, however, so this evidence is usually lost, and parents learn about difficulties only from grades. Advances in automatic speech recognition (ASR) and large language models (LLMs) make it possible to turn spoken lessons into structured data and analyze them automatically [1]–[3].

This review informs the design of such a system, which we are developing for online schools, tutoring centers, and independent tutors in Kazakhstan. A teacher uploads the recording of a one-to-one or small-group lesson held in Kazakh or Russian. The system cleans the audio, separates teacher and student speech, transcribes it, and passes an anonymized transcript to an LLM. The LLM identifies the topics of the lesson and rates the student's understanding of each topic with time-coded evidence from the dialogue. These ratings, combined with the results of short follow-up quizzes, update a per-student knowledge map. The system then produces separate reports for the teacher, the student, and the parent.

Each stage of this pipeline can fail in ways that later stages inherit. A misrecognized word or an answer attributed to the wrong speaker changes what the LLM sees; an LLM that agrees with teachers on broad categories may still misjudge a specific answer; and a report that teachers do not trust will not be used. We therefore structure the review around five questions:

- **RQ1:** How reliable are ASR and speaker diarization as a source of educational evidence, and how does reliability depend on task and language?
- **RQ2:** How well can LLMs analyze classroom dialogue and judge students' answers compared with human experts?
- **RQ3:** How can repeated evidence be turned into an estimate of a student's knowledge state?
- **RQ4:** What does the literature say about trust, human oversight, and fairness in teacher-facing AI?
- **RQ5:** Which gaps remain for an end-to-end system of this kind?

Section II describes the review method. Sections III–VI answer RQ1–RQ4. Section VII compares the studies, identifies research gaps (RQ5), and derives implications for the proposed system. Section VIII concludes.

II. Review Method

This is a structured narrative review. It is purposive rather than exhaustive: the aim was a compact evidence base covering every stage of the planned system. Searches were run in 2026 in Google Scholar and directly on Frontiers, npj, Wiley Online Library, ScienceDirect, Zenodo, MDPI, and the Scientific Journal of Astana IT University. Search terms combined "classroom dialogue analysis," "large language models education," "automatic speech recognition education," "speaker diarization classroom," "Kazakh speech recognition," "automated short answer grading," "knowledge tracing," "learning analytics," and "explainable AI education." A first round produced 20 articles; a second round, run after the language gap had been identified, added three studies on Kazakh and Russian ASR and on LLM-based answer grading.

Articles were included if they were (I1) peer-reviewed journal articles, (I2) connected to at least one stage of the system,

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(I3) empirical or methodological studies or reviews, and (I4) written in English. Conference papers, books, and preprints were excluded. The final set contains 23 articles: 13 research studies (Table I) and 10 reviews (Table II). Twenty-one were published in 2022–2026; two older reviews [4], [5] were kept as foundational maps of the field. Each research study was appraised qualitatively for transparency of method and data, comparison with a human or baseline reference, treatment of fairness, and transferability to our setting. Numerical results are quoted as reported by the original authors.

III. Speech Technologies as Evidence (RQ1)

The speech studies agree that spoken interaction can become machine-readable evidence, but they also show that accuracy depends on the task, the speakers, and the language. Petscher et al. [6] compared a commercial speech-verification system with human raters for 429 kindergarten children. At the most liberal threshold, item-level agreement ranged from poor ($\kappa = 0.15$, phoneme blending) to substantial ($\kappa = 0.70$, expressive vocabulary), fell as thresholds tightened, and differed between racial subgroups; human raters themselves showed larger group differences on some tasks. Because speech verification checks a known target word, these results transfer only indirectly to open dialogue, but they show that one aggregate accuracy figure hides large variation. Bailey et al. [7] reach a similar conclusion from the side of measurement theory: ASR and NLP systems trained on data that under-represent some English varieties produce less valid scores for those speakers, so recognition accuracy is part of assessment validity.

Speaker attribution is a separate source of error. Optimizing diarization for noisy classroom group work, Wang et al. [8] reached a diarization error rate (DER) of about 34%, a correlation of 0.62 between estimated and annotated student speaking time, and 69.17% accuracy in distinguishing teacher from child speech, with no significant accuracy differences by skin tone or gender. Their best configuration combined ECAPA-TDNN speaker embeddings with Whisper. One-to-one lessons, the main use case of our system, should be easier than group discussions, but the result warns that teacher and student turns cannot be assumed to be separated correctly.

Two studies from Kazakhstan address the languages of our system directly. Makhataeva et al. [9] evaluated Whisper large-v3-turbo fine-tuned by the Institute of Smart Systems and Artificial Intelligence (ISSAI). On benchmark data, fine-tuning reduced Kazakh WER from 21.55% to 8.84% while Russian stayed close to its baseline (6.15% vs. 5.89%). Yet when 38 adults read prepared passages aloud, system WER was 26.32% for Kazakh, 11.86% for Russian, and 9.65% for English, with significant differences between Kazakh and the other languages. Mussakhojayeva et al. [10] trained 22 ASR models for ten Turkic languages and found that multilingual training on related languages reduced CER and WER by 56.7% and 54.3% on average, more than adding Russian or English data. The same model, however, reached a Kazakh WER of 6.0% on one test corpus and 29.0% on another. Both studies use read or crowd-sourced speech, not lessons.

Two further studies place these findings in context. The meta-analysis of Lou et al. [17] reports that the effect of ASR on second-language listening depends on how advanced the technology is, and the multimodal model of Liu et al. [11] for assessing teachers' classroom language found in ablations that the text modality contributed most. Liu et al. used sentiment benchmarks rather than classroom data for most of their evaluation, and reached only 66.1% four-class accuracy on their own teacher recordings, so the finding is indicative rather than conclusive.

The methodological quality of these studies is uneven, and none matches our setting fully. Petscher et al. have the strongest design, with a large sample, item-level analysis, and subgroup comparisons, but study single words read by young children. Wang et al. use spontaneous classroom audio, the most realistic data in the set, but group discussions rather than teacher-led question-and-answer exchanges. Makhataeva et al. compare languages within the same participants, which controls for speaker differences, but draw all 38 participants from one university and use prepared texts. Mussakhojayeva et al. compare many models systematically with open data and code, but their models are not Whisper-based and their test sets are not conversational. Together, the evidence supports a text-first pipeline, but it also shows that published WER figures say little about accuracy on a new kind of recording. An ASR transcript should be treated as an intermediate result whose errors propagate, and its accuracy must be measured on the target data, separately for each language.

IV. LLMs for Dialogue Analysis and Answer Judgment (RQ2)

Two studies tested LLMs as coders of classroom dialogue. Long et al. [1] compared a customized GPT-4 model with expert coding of middle-school mathematics and Chinese lessons and report large time savings and high consistency, with disagreement concentrated in particular codes. Xu et al. [2] coded more than 400 Chinese talk turns from online mathematics lessons. Agreement among the human standard and two models was moderate at the category level (Fleiss's $\kappa = 0.46$); against the human standard, GPT-4o reached Cohen's $\kappa = 0.69$ and the GPT-3.5-based ChatGPT 0.33, while performance on fine-grained codes was unsatisfactory for both. The authors therefore propose that AI handle category-level coding and humans refine details.

Discourse codes are not the same as knowledge, however. Closer to our task, Kortemeyer [12] asked GPT-4, without task-specific training, to grade 14,186 written student answers to 504 science and electronics items. In the 2-way scheme, F1 was 0.744 on SciEntsBank and 0.611 on Beetle, and 0.731 and 0.651 when no reference answer was given. This is comparable to older hand-engineered grading systems but below fine-tuned BERT-family models. The author notes that sending student data to a cloud model raises privacy concerns and that a local open model (Llama 2) performed worse in preliminary tests.

These three studies are difficult to compare directly. They differ in model generation (GPT-3.5 to GPT-4o), task (discourse coding versus answer grading), subject, and language, and none of them tests how stable the output is across prompts or repeated runs; Kortemeyer reports a single run. Xu et

TABLE I
RESEARCH STUDIES INCLUDED IN THE REVIEW

#	Study	Data / context	Method	Key result (as reported)
1	Petscher et al. [6]	429 kindergarten students, USA; three word-level tasks	Commercial speech verification vs. human raters; Cohen's κ; Black/White subgroups	Item-level κ 0.15 (blending), 0.70 (vocabulary), 0.52 (word reading)
2	Bailey et al. [7]	Child language and literacy assessment, under-served students	Validity framing (Interpretation/Use Argument); ASR + NLP experiments	Under-represented English varieties in training data harm scoring validity
3	Wang et al. [8]	Noisy classroom group discussions	ECAPA-TDNN embeddings + Whisper; design-choice comparison	DER $\approx$ 34%; r = 0.62 for student talk time; teacher/child accuracy 69.17%
4	Makhataeva et al. [9]	38 adults; read Kazakh, Russian, English passages	Whisper large-v3-turbo fine-tuned by ISSAI; WER/CER; ANOVA	Benchmark Kazakh WER 21.55% $\rightarrow$ 8.84%; user-study WER 26.32% (kk) vs. 11.86% (ru)
5	Mussakhojayeva et al. [10]	Ten Turkic languages; Common Voice, KSC; Russian/English controls	13 monolingual vs. 9 multilingual conformer models	Mean CER/WER reduction 56.7%/54.3%; same model: Kazakh WER 6.0% (KSC) vs. 29.0% (Common Voice)
6	Liu et al. [11]	CMU-MOSI/MOSEI; self-built teacher data	BMRN multimodal (text, audio, video) fusion	MOSEI MAE 0.547, r = 0.781; 66.1% four-class accuracy; text modality contributes most
7	Long et al. [1]	Middle-school mathematics and Chinese lessons	Customized GPT-4 vs. expert coders	Large time savings; high consistency, disagreement in specific codes
8	Xu et al. [2]	400+ Chinese talk turns, online mathematics	GPT-3.5-based ChatGPT and GPT-4o vs. human standard	Category level: κ = 0.69 (GPT-4o), 0.33 (ChatGPT); code level unsatisfactory
9	Kortemeyer [12]	SciEntsBank, Beetle; 504 items, 14,186 answers	Out-of-the-box GPT-4, with/without reference answers	2-way F1 0.744 and 0.611; comparable to hand-engineered, below fine-tuned models
10	Sarsa et al. [13]	Public and programming-course datasets	DKT, LSTM-DKT, DKVMN, SAKT vs. simple baselines	Gains of deep models inconsistent; sensitive to hyperparameters, metrics, seeds
11	Cukurova et al. [14]	One-to-one online tutoring sessions	CM-SPAM sequence mining; J48, JRIP classifiers	Tutor-behavior sequences separate effective from weaker sessions
12	Aldino et al. [15]	21 higher-education teachers	BERT classifier + GPT-4o mini (PolyFeed); logs, interviews	ML suggestions adopted; LLM text heavily revised by teachers
13	Wang et al. [16]	59 pre-service teachers (30 vs. 29)	AI dialogue analysis with vs. without explanations	Higher trust and acceptance with explanations; no added cognitive load

al. use authentic but modest data from one subject and one language. All three work with clean text: none feeds the LLM an ASR transcript, so the effect of recognition errors on LLM judgments remains untested.

The reviews describe the same promise and the same limits. Wang, Tao, and Chen [3] map 68 studies of AI in classroom discourse and call for evaluation of practical effects as deep learning and LLMs spread. Yan et al. [18], reviewing 118 papers on LLM-supported educational tasks, identify practicality, reliability, and ethics as the main concerns, and Sharma et al. [19] find that evidence of educational impact of LLMs in personalized learning is still thin. For our system, the evidence suggests three rules: the LLM should rate understanding against explicit, coarse categories with examples; its ratings should be validated against teachers in each subject; and the choice between a cloud and a local model is a trade-off between accuracy and data protection.

TABLE II
REVIEW PAPERS INCLUDED IN THE REVIEW

Review	Coverage	Main point used here
Zhai et al. [5]	100 papers, 2010–2020	Layers of AI applications; ethics, changing roles
Zawacki-Richter et al. [4]	AI in higher education	Educators under-represented in AI research
Chiu et al. [20]	2012–2021	13 AI roles, 7 outcomes; field scattered
Wang et al. [3]	68 studies, one decade	AI in classroom discourse; test practical effects
Yan et al. [18]	118 papers	LLM tasks; practicality, reliability, ethics
Sharma et al. [19]	LLMs in personalized learning	Thin evidence of educational impact
Abdelrahman et al. [21]	Knowledge tracing survey	Bayesian to deep models; open questions
González-Pérez et al. [22]	36 studies, 2021–2025	Link learning-analytics models to decisions
Boujmiraz et al. [23]	Performance prediction	XAI adopted less than prediction
Lou et al. [17]	Bibliometrics + meta-analysis	ASR effect depends on technology maturity

V. FROM ANSWERS TO A KNOWLEDGE STATE (RQ3)

A rating of one answer is not yet an estimate of what a student knows. Knowledge tracing addresses this by updating a learner model from sequences of evidence; the survey of Abdelrahman et al. [21] traces the field from Bayesian knowledge tracing to memory- and attention-based deep models. Sarsa et al. [13], however, show that deep models such as DKT, DKVMN, and SAKT outperform simple baselines only inconsistently and that reported gains depend on hyperparameters, metrics, random seeds, and hardware. Cukurova et al. [14] show that sequences of tutor behavior in one-to-one online tutoring, mined with CM-SPAM and classified with decision trees, separate effective from weaker sessions; the target is tutoring quality, but the study confirms that interaction sequences carry more signal than isolated utterances. Reviews of learning analytics [22] and performance prediction [23] add that models are useful only when tied to educational decisions, and that explainability is adopted less often than prediction.

All of this work uses structured logs rather than spoken dialogue. For our system, the evidence suggests a transparent,

rule-based aggregation of dialogue ratings and quiz results into a per-topic index as the first model, with any learned knowledge-tracing model compared against it as a baseline rather than assumed to be better.

VI. Trust, Oversight, and Fairness (RQ4)

Teacher-facing AI is adopted only if teachers can check it. Aldino et al. [15] observed 21 higher-education teachers using a feedback tool that combined a BERT classifier with GPT-4o mini: teachers often accepted the classifier's suggestions but heavily revised the LLM's text, citing tone and trust. In a randomized study with 59 pre-service teachers, Wang et al. [16] found that adding explanations to AI analysis of classroom dialogue significantly increased trust and technology acceptance without increasing cognitive load. Both studies are small, and both measure perceptions over a short period: Aldino et al. studied feedback writing rather than assessment, and the participants of Wang et al. were pre-service teachers rather than practicing ones. Whether explanations keep teachers engaged over months of real use is unknown. The broad reviews point the same way: Zawacki-Richter et al. [4] find educators under-represented in AI research, Zhai et al. [5] warn about inappropriate use and changing teacher roles, and Chiu et al. [20] describe a scattered field in which links between AI technologies and learning outcomes are unclear.

Fairness is the other side of trust. The speech studies show that accuracy must be reported per task and per group [6], [7], that such checks are feasible [8], and that language is one of the groups that matter [9]. A system used with both Kazakh- and Russian-speaking students that worked well for only one language would treat students unequally.

VII. Discussion

A. Comparison of the Studies

Table III shows which stages of the pipeline each research study addresses. No study combines diarization, ASR, dialogue analysis, and knowledge assessment; the broadest [7] covers four of six components without speaker separation or explanations, and the two studies on Kazakh and Russian ASR [9], [10] do not concern education. The studies also measure different things with different metrics: Cohen's and Fleiss's κ, F1, WER, CER, DER, correlations, MAE, and perception scales, applied to objects from single words read by children to lessons of adults. Human raters are not error-free either [6]. Published figures are therefore orientation points rather than targets, and component-level and end-to-end results must be reported separately.

TABLE III
Pipeline components addressed by the research studies

Study	ASR	DIA	DLG	KNW	EXP	FAI
Petscher [6]	✓			✓		✓
Bailey [7]	✓		✓	✓		✓
Wang (diar.) [8]	✓	✓				✓
Makhataeva [9]	✓					✓
Mussakhojayeva [10]	✓					
Liu [11]			✓			
Long [1]			✓			
Xu [2]			✓			
Kortemeyer [12]			✓	✓		
Sarsa [13]				✓		
Cukurova [14]			✓			
Aldino [15]			✓		✓	
Wang (XAI) [16]			✓		✓	
Total	5	1	8	4	2	4

ASR: speech recognition or verification; DIA: diarization; DLG: dialogue or text analysis; KNW: assessment of knowledge or learning state; EXP: explainability or human oversight; FAI: fairness, validity, or per-group (e.g., per-language) analysis. A mark means a central focus of the study (authors' classification).

B. Research Gaps (RQ5)

Six gaps follow from the comparison. (1) *Integration:* the components have not been combined into one teacher-facing workflow. (2) *Validation against knowledge:* LLM studies validate against discourse categories [1], [2] or written benchmark answers [12], not against teachers' judgments of what a student understood in a spoken lesson. (3) *Error propagation:* recognition, diarization, and fairness errors are documented [6]–[8], but not how they change an LLM's assessment. (4) *Explainability:* explanations raise trust in AI labels of dialogue [16], but how to explain an assessment of a student's understanding has not been studied in the reviewed set. (5) *Longitudinal assessment:* knowledge tracing is built on structured logs [13], [21], and linking it to conversational evidence remains open. (6) *Language:* the educational evidence comes mainly from English and Chinese settings [1], [2], [6], [7], [11]. The only Kazakh and Russian evaluations [9], [10] use read or crowd-sourced monolingual speech; to the best of our reading, no reviewed study evaluates spontaneous Kazakh or Russian teacher–student dialogue, Kazakh–Russian code-switching, or diarization in these languages, even though Kazakh lags Russian by a wide margin under favorable conditions [9].

C. Implications for the Proposed System

Fig. 1 maps the reviewed evidence onto the pipeline of our system. Because lessons in Kazakhstan are taught in both Kazakh and Russian, the system should support both languages from the start. The evidence shows that this requires a Kazakh-adapted recognition model, separate accuracy targets for each language, and code-switched speech treated as a distinct evaluation condition rather than assumed to work [9], [10]. Speaker turns should be checked by the teacher before analysis [8]; the LLM should rate understanding against explicit categories and cite the time-coded fragments on which each rating rests [2], [16]; and the knowledge index should be computed by transparent rules rather than by the LLM, so that results are reproducible [13]. Evaluation should separate component metrics from end-to-end outcomes:

- *ASR and diarization:* WER and DER on the project's own lesson recordings, reported separately for Kazakh, Russian, and code-switched speech and for each recording condition [8]–[10].
- *LLM analysis:* agreement (κ or F1) with expert teachers' topic ratings, comparing a cloud and a locally hosted model on accuracy and data protection [2], [12].

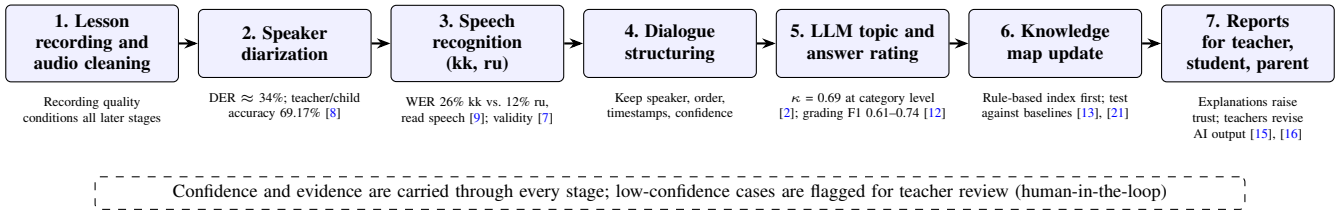


Fig. 1. Pipeline of the proposed system and the literature evidence relevant to each stage.

- *End-to-end assessment:* agreement between the system's topic status and the judgments of at least two teachers, with inter-rater agreement reported [6] and a simple baseline for comparison [13].
- *Trust and usability:* a teacher study with and without explanations [16].

Throughout, the system should support rather than replace the teacher's judgment [4], [15].

D. Limitations of This Review

The review is purposive and covers 23 articles, so other search terms might have retrieved other studies; three articles were added in a second, targeted round, which may favor sources that confirm the gaps found in the first. Conference papers and preprints were excluded, so the review lags behind the fastest-moving ASR and LLM research. Quality appraisal was qualitative and done without a second independent rater. Results for specific models (GPT-4, GPT-4o, Whisper large-v3-turbo) may not hold for newer ones, the only Kazakh and Russian evidence concerns read speech outside the classroom, and all figures are quoted from the original articles rather than reproduced.

VIII. Conclusion

The reviewed literature supports the feasibility of each stage of the proposed system but not yet of the system as a whole. Speech technologies turn lessons into analyzable text, but their accuracy depends on task, recording conditions, and language, and Kazakh is recognized far less accurately than Russian even under favorable conditions (RQ1). LLMs reduce the effort of dialogue analysis and can judge short answers with moderate accuracy, but agreement with experts drops for fine-grained judgments (RQ2). Knowledge tracing offers ways to aggregate evidence over time, but complex models do not reliably beat simple baselines (RQ3). Explanations and human oversight are decisive for teacher acceptance, and accuracy must be checked per group and per language (RQ4). Six gaps remain: integration, validation against subject knowledge, error propagation, grounded explanations, longitudinal assessment, and evidence for spontaneous Kazakh, Russian, and code-switched dialogue (RQ5). The proposed system addresses these gaps by combining speaker-aware Kazakh and Russian ASR, evidence-based LLM ratings, and a transparent knowledge index in a teacher-facing workflow, and by evaluating each stage against teachers' judgments.

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